Development of a Web Application for Textile Workshop Management

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Introduction

The textile industry plays an important role in many local economies. Small and medium textile workshops produce garments such as T-shirts, dresses, uniforms, and other products.

However, many workshops still manage their operations manually using notebooks or spreadsheets. This leads to several problems:

- Poor order tracking
- Inaccurate inventory management
- Manual salary calculations
- Lack of production analytics

Goal: Develop a web application that automates these processes and improves workshop management.

Problem Statement

Many textile workshops face operational challenges:

- Orders are tracked manually
- Materials are not properly monitored
- Production reports are recorded on paper
- Salary calculations are performed manually
- Management lacks clear analytical insights

These problems reduce efficiency and increase the risk of errors.

A digital management system can significantly improve production control and business transparency.

Research and Existing Solutions

ERP Systems

SAP • Oracle ERP • Microsoft Dynamics

Advantages:

Comprehensive business management

Disadvantages:

- ✗ Complex implementation
- ✗ Expensive
- ✗ Not suitable for small workshops

Manufacturing Execution Systems (MES)

Advantages:

Real-time production monitoring

Disadvantages:

- ✗ Designed for large factories
- ✗ Difficult to adapt for small textile businesses

Therefore, a simpler and more specialized solution is needed.

Proposed Solution

This project proposes a Web Application for Textile Workshop Management.

The system will provide a centralized platform for managing:

- **Orders**
- **Inventory**
- **Production**
- **Employees**
- **Financial Data**

The application will help workshop administrators track operations in real time and improve decision-making.

Main Features of the System

Authentication

- Login and password
- Password recovery
- Two-factor authentication

Dashboard

- Monthly revenue
- Active orders
- Completed orders
- Defective items
- Inventory overview

Clients

- Client database
- Order history

Orders

- Order creation
- Product types
- Auto price calculation
- Order status tracking

Production and Inventory Management

Inventory Management

- Material tracking
- Purchase operations
- Stock updates
- Usage history

Technical Card (Bill of Materials)

Each product has a list of required materials.

Example — T-shirt production requires:

- Fabric
- Thread
- Buttons

Materials are automatically deducted from inventory during production.

Employee and Salary Management

Employee Data

- Name
- Position
- Phone number
- Piecework rate
- Hiring date

Salary Calculation

Salary calculation is automated:

Salary = produced items × piecework rate

The system generates monthly salary reports for employees.

System Architecture

Frontend	Backend	Database	File Storage	Deployment
React	Java Spring Boot	PostgreSQL	MinIO (S3-compatible)	Docker containers

This architecture ensures scalability, reliability, and easy deployment.

MVP (Minimum Viable Product)

MVP Features

- ✓ Authentication system
- ✓ Dashboard
- ✓ Client management
- ✓ Order management
- ✓ Inventory tracking
- ✓ Production reporting

Future Versions

- Advanced analytics
- Mobile support
- Integration with accounting systems
- Multi-workshop management

Expected Results

- Improved production control
- Accurate inventory tracking
- Automated salary calculation
- Better financial transparency
- Faster order processing

The application will help textile workshop owners manage their operations more efficiently.

References

1 Monk, E., & Wagner, B. (2013). Concepts in Enterprise Resource Planning. Cengage Learning.

2 Laudon, K., & Laudon, J. (2020). Management Information Systems. Pearson.

3 Kletti, J. (2007). Manufacturing Execution Systems. Springer.

Additional sources were analyzed to study manufacturing management systems and small business digitalization.

Thank you for your attention!